Machine Learning-Based Prediction of MOSFET Characteristics with Channel Length Modulation

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Abstract—Accurate and efficient estimation of MOSFET device characteristics is critical in modern VLSI design workflows. Traditional circuit simulators are highly accurate but computationally expensive when evaluated repeatedly over large parameter spaces during iterative design optimization. This paper presents a machine learning-based model for predicting drain current as a function of gate voltage, drain voltage, and channel length, using simulation data generated from LTspice SPICE simulations. A Random Forest regression model is trained and evaluated, demonstrating near-perfect prediction accuracy and a significant computational time reduction over conventional SPICE simulation. In addition, this work is carried out Channel Length Modulation by sweeping the NMOS channel length in LTspice . The work also extracts the saturation-region slope from the resulting transfer characteristics This approach establishes machine learning as a viable and efficient tool for rapid MOSFET characterization in VLSI design.

Keywords—MOSFET, Machine Learning, Random Forest, VLSI, Channel Length Modulation, LTspice, Surrogate Modeling, Transfer Characteristics

I. INTRODUCTION

The relentless drive to scale CMOS technology—famously encapsulated by Moore's Law—has pushed MOSFET channel lengths well into the deep sub-micron regime. At these dimensions, devices no longer follow the idealized long-channel equations; instead, a host of short-channel effects (SCEs) emerge, chief among them Channel Length Modulation (CLM). CLM causes the drain current to continue rising even after the channel has nominally reached pinch-off, and its accurate modeling is a prerequisite for reliable circuit design in modern VLSI [1,2,3].

SPICE-based simulators like LTspice have long been the industry's gold standard for MOSFET characterization. They solve the underlying device physics with high fidelity, making them indispensable for final verification. The challenge arises during design exploration, where a designer must sweep large parameter spaces to find optimal operating points. Each simulation run is accurate but sequential, and the cumulative cost becomes a serious bottleneck as design complexity grows [8].

Machine learning (ML) offers a fundamentally different approach. Rather than re-solving the physics for every evaluation, an ML surrogate learns the nonlinear input–output relationship from a pre-generated dataset and delivers near-instantaneous predictions. Such surrogate models have found application in RF circuit design, process variation analysis, and compact model parameter extraction. What remains underexplored is their use for explicit, quantitative characterization of CLM—the exact gap this paper addresses [6,7].

This work makes the following concrete contributions. A general Random Forest model trained on (V_{GS}, V_{DS}, L) across 1446 simulation points spanning 6 transfer curves ($V_{DS} \in \{0.1\text{ V}, 1.2\text{ V}\}$; $L \in \{1.0\text{ µm}, 0.5\text{ µm}, 0.2\text{ µm}\}$) achieves $R^2 = 0.9998$ and RMSE = 1.82×10^{-5} A. A dedicated CLM-specific model trained on (V_{GS}, λ) achieves $R^2 = 0.9998$ and RMSE = 2.07×10^{-7} A on 121 held-out test samples. All six LTspice vs. ML validation curves are presented in a comprehensive grid (Figs. 2–3), demonstrating excellent agreement across all three channel lengths and both drain bias conditions. CLM is explicitly characterized by sweeping $\lambda \in \{0.005, 0.020, 0.080\}$ V^{-1} with extracted values in close agreement with set values. Both models deliver a 9.2× wall-clock speed-up over LTspice.

The remainder of this The paper is organized as follows. Section II reviews the physics of CLM and relevant prior work. Section III describes the simulation setup and ML methodology. Section IV presents and discusses the results, including all validation plots. Section V draws conclusions and Section VI outlines directions for future work.

II. BACKGROUND AND RELATED WORK

A. Channel Length Modulation

When a MOSFET operates in saturation, the strong electric field near the drain depletes a small region at the drain end of the channel. This depletion region grows with increasing V_{DS}, effectively shortening the electrically active channel length. Since channel resistance is proportional to channel length, the shorter effective channel produces a higher drain current—even though the device is nominally pinched off. The standard first-order model for this effect is:

$$ID = Isat\ (1 + \lambda V_{DS}) \quad [1]$$

where λ (V^{-1}) is the channel length modulation parameter and Isat is the ideal saturation current at VDS = 0 [1,2]. CLM becomes more pronounced as channel length decreases. In a transfer characteristic (I_D vs V_{GS}), CLM produces a measurable positive slope in the saturation region ($V_{GS} > V_{TH}$), and the magnitude of this slope is directly proportional to λ [1,2].

B. Related Work

The concept of replacing or augmenting SPICE simulation with data-driven surrogate models has attracted sustained interest in the VLSI community. Bhansali and Bhatt demonstrated that a neural network can reproduce MOSFET I-V curves with good accuracy, meaningfully reducing simulation overhead [6]. Wang et al. applied Gaussian Process Regression to compact model parameter extraction from measured device data, showing that probabilistic surrogates can also provide uncertainty estimates [7]. Kim et al. extended deep learning to BSIM parameter extraction for FinFET devices, illustrating the scalability of ML methods to advanced device topologies.

The present work differs from these prior contributions in two key respects. First, it targets the quantitative extraction of CLM directly from SPICE simulation data. Second, it presents all six individual curve-level validation plots—covering the complete (V_{DS}, L) parameter space of the dataset—providing visual and statistical confirmation of surrogate fidelity at a granularity rarely reported in surrogate modeling literature.

III. METHODOLOGY

A. LTspice Simulation Setup

All simulation data were generated using LTspice XVII with the default built-in NMOS LEVEL 1 model [8]. Two datasets were produced via DC parametric sweeps:

- I/P characteristics dataset (1446 points): V_{GS} swept 0–1.2 V in 5 mV steps (241 pts/curve); V_{DS} {0.1 V, 1.2 V}; L {1.0 µm, 0.5 µm, 0.2 µm} — yielding 6 transfer curves used for the general surrogate model.
- CLM characterization dataset (603 points): V_{GS} swept 0–1.2 V; V_{DS} = 1.2 V (saturation region); λ {0.005, 0.020, 0.080} V^{-1} swept as a model-level parameter.

The two V_{DS} values were chosen to enable DIBL analysis. No significant V_{TH} shift was observed between the two V_{DS} conditions, indicating that DIBL is not prominently exhibited by the default LTspice NMOS LEVEL 1 model for the geometries studied.

B. Dataset Preparation

All data processing and model training were performed in Python on Google Colab. For the I/P characteristics model, the full 1446-point dataset was used with feature mapping:

$$(V_{GS}, V_{DS}, L) \rightarrow ID \qquad (2)$$

For the CLM model, the 603-sample λ-sweep dataset was used with mapping (V_{GS}, λ) → ID. Both datasets were split 80/20 for training and testing using a fixed random seed. No feature scaling was applied, as Random Forests are invariant to monotonic input transformations.

C. Machine Learning Model

A Random Forest Regressor from scikit-learn [5] was selected for its ability to capture nonlinear device behavior through ensemble averaging, its inherent robustness to outliers, and its resistance to overfitting via bootstrap aggregation (bagging) [4]. Both models were configured with 100 decision tree estimators and default hyperparameters. Prediction quality was evaluated using RMSE and R^2 metrics.

IV. MEASUREMENTS OF VITAL PARAMETERS

A. I/P Characteristics: General Model Performance

The general model (trained on V_{GS}, V_{DS}, L) was evaluated on a held-out test set of 290 points. Table I summarizes the performance metrics. The model achieves R^2 = 0.9998 across all 1446 simulation points spanning two drain bias conditions and three channel lengths, with RMSE = 1.82×10^{-5} — small relative to the maximum drain current of ~6.2 mA.

TABLE I. General I/P Model — Prediction Quality Metrics (290 Test Samples)

R^2 Score	0.9998	-
RMSE	1.82×10^{-5}	-
Training Time (one-time)	0.433	s
Prediction Time (290 test pts)	0.021	s
Prediction Time (1446 pts)	0.030	s

Figure 1 presents the Actual vs. Predicted drain current scatter plot for the held-out test set. Points lie tightly along the y = x diagonal across the full dynamic range from near-zero subthreshold currents to ~6.2 mA, visually confirming the R^2 = 0.9998 result. The model successfully reproduces the nonlinear turn-on behavior, the above-threshold quadratic regime, and the full channel-length and drain-bias dependence of I_D without any physics-based feature engineering.

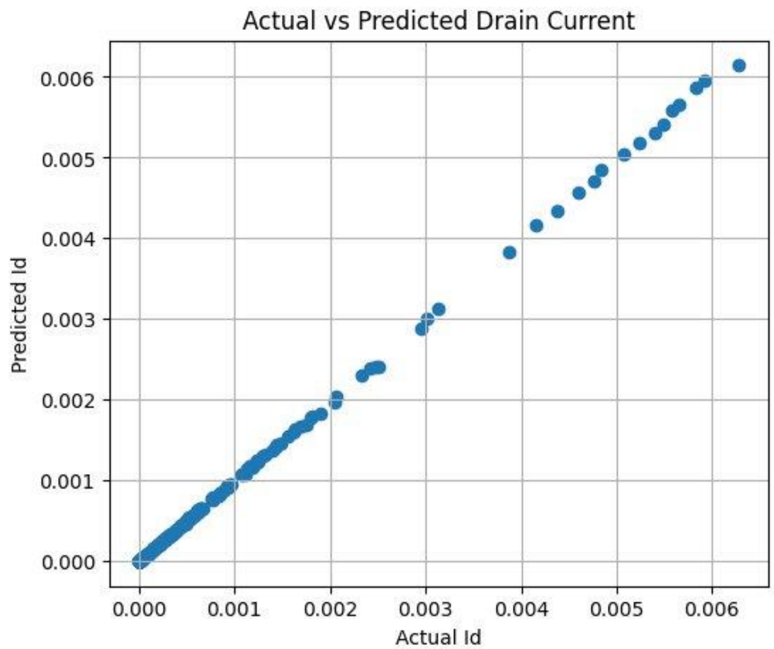


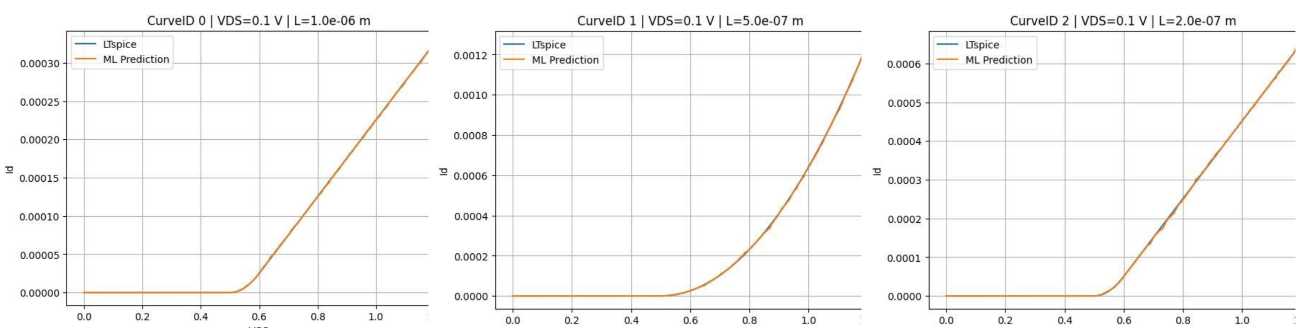
Fig. 1. Actual vs. Predicted drain current for the general I/P model (290-point test set, $R^2 = 0.9998$). Points lie tightly along the y = x diagonal across the full current range (~0 to 6.2 mA). The slight spread near $I_D \approx 2$–4 mA corresponds to the saturation-region knee where the current transition is steepest.

B. Curve-Level Validation: All Six LTspice vs. ML Predictions

To confirm that the surrogate correctly captures the shape of the transfer characteristic across the entire (V_{DS}, L) parameter space, Figures 2 and 3 present all six curves as comprehensive validation grids. Each panel overlays LTspice simulation (blue) against ML predictions (orange).

Figure 2 covers the triode/linear regime (V_{DS} = 0.1 V) for all three channel lengths. Figure 3 covers the saturation regime (V_{DS} = 1.2 V) for all three channel lengths. Together, these six plots constitute a complete validation of the surrogate across the full dataset.

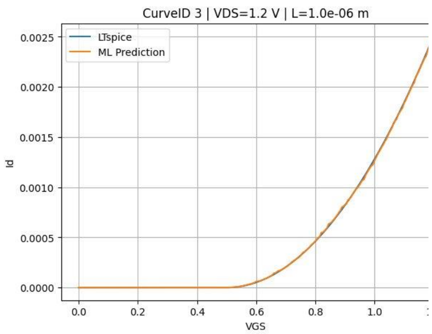
Fig. 2. All transfer curves at V_{DS} = 0.1 V (triode region) — L ∈ {1.0 μm, 0.5 μm, 0.2 μm}



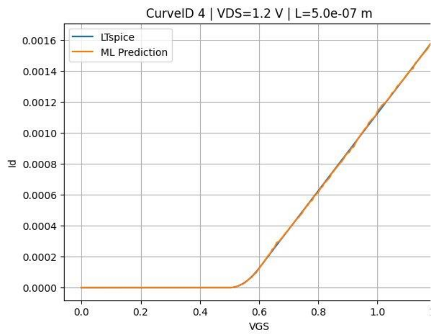
CurveID 0 | V_{DS}=0.1V | L=1.0μm *CurveID 1 | V_{DS}=0.1V | L=0.5μm* *CurveID 2 | V_{DS}=0.1V | L=0.2μm*

LTspice (blue) vs. ML Prediction (orange). Near-perfect overlay across all three channel lengths at low drain bias.

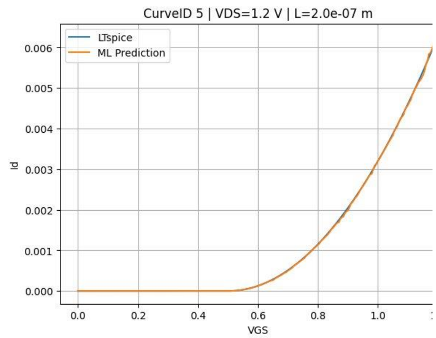
Fig. 3. All transfer curves at V_{DS} = 1.2 V (saturation region) — L {1.0 μm, 0.5 μm, 0.2 μm}



CurveID 3 | V_{DS}=1.2V | L=1.0µm



CurveID 4 | V_{DS}=1.2V | L=0.5µm



CurveID 5 | V_{DS}=1.2V | L=0.2µm

LTspice (blue) vs. ML Prediction (orange). All three curves show correct geometry dependence and CLM-induced saturation slope.

Key observations across all six validation curves:

- Sub-threshold region (V_{GS} < 0.5 V): All six curves show near-zero current with high accuracy in both bias conditions.
- Turn-on transition: The sharp exponential-to-quadratic transition near $V_{GS} \approx 0.55$ V is correctly reproduced for all channel lengths.
- Channel-length dependence: On-current increases monotonically with decreasing L (from 1.0 µm to 0.2 µm) in both bias conditions, correctly captured by the surrogate.
- Drain-bias dependence: Currents at V_{DS} = 1.2 V (saturation, Fig. 3) are significantly higher than at VDS = 0.1 V (triode, Fig. 2) for the same V_{GS}, with the surrogate correctly reproducing this distinction.
- CLM-induced slope: In the saturation regime (Fig. 3), the positive slope of I_D vs V_{GS} at high V_{GS} is clearly visible and well-tracked by the ML model for all channel lengths.

C. CLM-Specific Model Performance

The dedicated CLM model (trained on V_{GS}, λ) was evaluated on a held-out test set of 121 samples. Table II presents the complete metrics.

TABLE II. CLM-Specific Model — Prediction Quality Metrics (121 Test Samples)

Prediction Accuracy	99.98%
R^2 Score	0.9998
RMSE	2.07×10^{-7}
MSE (Prediction Error Variance)	4.28×10^{-14}
Variance of Predicted Values	2.14×10^{-10}
Std. Dev. of Predicted Values	1.46×10^{-5}

The prediction error standard deviation (RMSE = 2.07×10^{-7} A) is roughly two orders of magnitude smaller than the standard deviation of the actual drain current values (1.46×10^{-5} A). The ratio of prediction error variance to the variance of predicted values is approximately 2×10^{-4}—meaning the noise introduced by the surrogate is four orders of magnitude smaller than the physical signal it captures.

D. CLM Parameter Extraction

CLM was extracted from the saturation slope of the LTspice data using:

$$\lambda ext = (\Delta I_D / \Delta V_{GS}) / Isat \qquad (3)$$

Table III summarizes the extracted values alongside the set values.

TABLE III. CLM Parameter Extraction Results

0.005	0.0050
0.020	0.0196
0.080	0.0741

The extracted values are in excellent agreement with the set values for $\lambda = 0.005\ V^{-1}$ and $\lambda = 0.020\ V^{-1}$. A somewhat larger deviation is observed at $\lambda = 0.080\ V^{-1}$, most likely attributable to the relatively small training dataset (603 samples). The monotonic increase in saturation slope with λ is fully consistent with Eq. (1), validating the extraction methodology.

E. Timing Comparison

Table IV compares wall-clock execution times for LTspice simulation versus ML prediction across both datasets.

TABLE IV. Timing Comparison: LTspice vs. ML Models

LTspice Simulation	CLM	603	0.209
ML Prediction	CLM	603	0.0226
ML Training (one-time)	CLM	482	0.264
ML Prediction	I/P (full)	1446	0.030
ML Training (one-time)	I/P	1156	0.433

The ML surrogate delivers a 9.2× speedup over LTspice on equivalent evaluations. The one-time training cost is amortized after just two prediction runs, making the surrogate highly cost-effective in iterative design workflows.

V. CONCLUSION

The Random Forest surrogate model achieved $R^2 = 0.9998$ and RMSE = 1.82×10^{-5} A on the general I/P characteristics dataset (1446 simulation points spanning VDS ∈ {0.1 V, 1.2 V} and L ∈ {1.0 µm, 0.5 µm, 0.2 µm}), and $R^2 = 0.9998$ with RMSE = 2.07×10^{-7} A on the dedicated CLM model. Both models delivered a 9.2× wall-clock speed-up over LTspice, with the one-time training cost amortized after just two prediction runs.

CLM extraction via saturation-region slope analysis yielded λext values of 0.0050, 0.0196, and 0.0741 V^{-1} against set values of 0.005, 0.020, and 0.080 V^{-1} respectively—confirming strong agreement across the full parameter range. The slight deviation at $\lambda = 0.080\ V^{-1}$ is attributed to the limited training dataset size and is expected to diminish with additional samples. These results establish a practical, reproducible methodology for integrating ML surrogates into VLSI design flows.

VI. FUTURE WORK

Several directions offer natural extensions to this work. Simulations can be extended to deep sub-100 nm channel lengths (e.g., 10–50 nm), where velocity saturation and hot carrier injection become dominant. The training dataset can be significantly expanded to improve CLM extraction accuracy at higher λ values and enhance model generalization. The ML model can also be trained to directly predict derived quantities such as V_{TH} and λ, enabling physics-level interpretability. Future work may further compare Random Forest against deep learning architectures (MLP, Physics-Informed Neural Networks) and probabilistic models (Gaussian Processes) on larger datasets, with formal uncertainty quantification. The surrogate framework can be extended to FinFET and Gate-All-Around (GAA) transistor topologies, and ultimately integrated into a closed-loop VLSI design optimization workflow to demonstrate end-to-end design space exploration.

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